BUSINESS DEVELOPMENT CAPSTONE REPORT

on

HimalayaInsight

AI-Powered Market Opportunity & Decision Intelligence Engine for Craft & Eco-Tourism MSMEs

Submitted in Partial Fulfilment of

AI Enabled Business Intelligence Internship

Internship Organization:

Technology Business Incubator (TBI)

Graphic Era (Deemed to be University)

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Executive Summary

The **HimalayaInsight** project was developed as part of the **AI Enabled Business Intelligence Internship** conducted by the **Technology Business Incubator (TBI), Graphic Era (Deemed to be University)**. The project focuses on demonstrating how Business

Intelligence (BI) techniques, combined with Artificial Intelligence (AI), can support data-driven decision-making for Micro, Small, and Medium Enterprises (MSMEs) operating in the craft, handicraft, and eco-tourism sectors.

The project was built using an e-commerce product dataset containing information such as product categories, brands, selling prices, market prices, customer ratings, and product descriptions. The data was cleaned and analysed using **Python (Pandas and Matplotlib)** and **SQL**, followed by the development of an interactive dashboard in **Tableau Public**. The dashboard presents key performance indicators (KPIs), pricing comparisons, category-wise performance, customer rating trends, and interactive filters that allow users to explore business insights efficiently.

Beyond traditional analytics, Generative AI tools were incorporated to support insight generation, prompt engineering, KPI interpretation, and business storytelling. Every AI-generated observation was carefully validated against the actual dataset to ensure accuracy and avoid unsupported conclusions. This approach demonstrated how AI can enhance analytical workflows while maintaining the integrity of evidence-based decision-making.

The analysis highlighted variations in pricing, customer engagement, and product performance across different categories, enabling the identification of opportunities for improved pricing strategies, product positioning, and market focus. Based on these findings, the project recommends adopting data-driven pricing decisions, continuously monitoring customer feedback, and integrating AI-assisted analytics into regular business operations. Overall, HimalayaInsight illustrates how Business Intelligence can transform raw business data into actionable insights that support strategic planning, operational improvement, and sustainable growth for MSMEs.

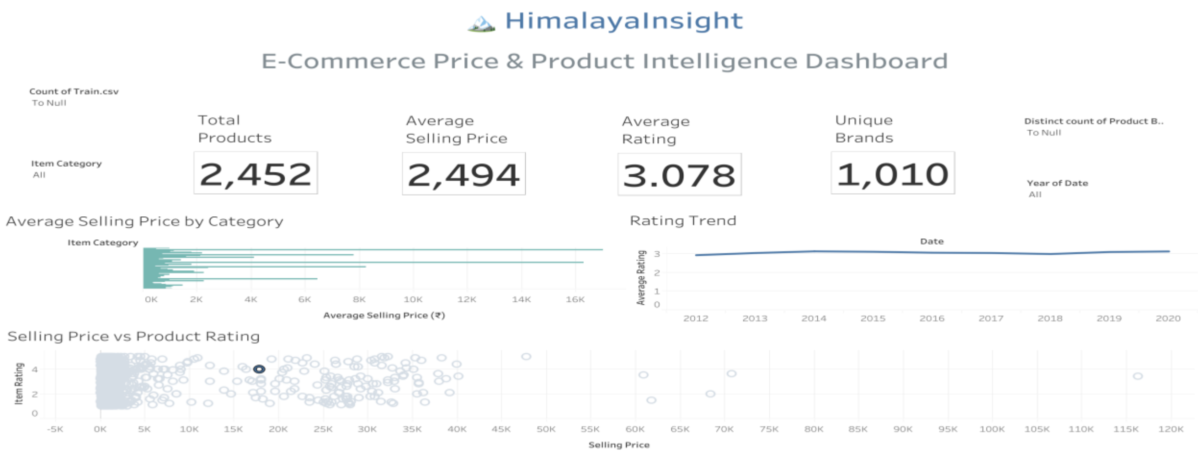

Figure 1: HimalayaInsight Business Intelligence Dashboard

Business Context

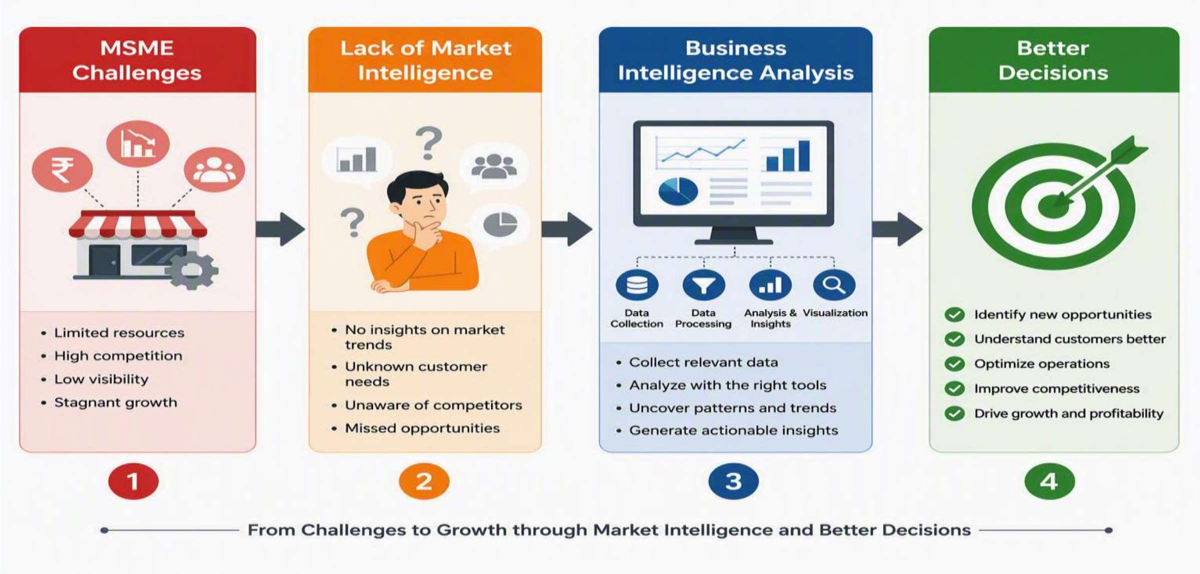

Figure 2. Business challenge addressed by HimalayaInsight.

India's craft, handicraft, and eco-tourism sectors play a significant role in supporting local economies, preserving cultural heritage, and generating employment opportunities for rural communities. A large portion of businesses operating in these sectors falls under the category of Micro, Small, and Medium Enterprises (MSMEs). Despite their economic importance, many of these enterprises continue to rely on experience-based decision-making rather than systematic market analysis.

One of the major challenges faced by MSMEs is the lack of access to structured market intelligence. Business owners often have limited visibility into pricing trends, customer preferences, competitor positioning, and product performance across digital marketplaces. As e-commerce platforms continue to expand, businesses are exposed to increasing competition, making it difficult to determine appropriate pricing strategies and identify profitable product categories.

The rapid growth of digital commerce and the increasing availability of data create an opportunity for MSMEs to adopt Business Intelligence (BI) practices. Affordable analytical tools, cloud-based dashboards, and AI-assisted insights can help small businesses interpret market information more effectively and make evidence-based decisions.

The HimalayaInsight project was conceived to address this gap by demonstrating how publicly available e-commerce data can be transformed into actionable business insights. The project focuses on analysing product categories, pricing patterns, customer ratings, and market distribution to support strategic decisions related to pricing, product positioning, and business development. The intended users of this solution include MSME owners, startup founders, business managers, and organizations involved in promoting local craft and tourism enterprises.

Market Analysis

3.1 Industry Overview

India's **handicraft, handloom, and eco-tourism sectors** represent an important segment of the country's MSME ecosystem. These industries preserve cultural heritage while contributing significantly to employment generation, rural livelihoods, and local economic development. Over the past decade, increasing internet penetration and the expansion of digital marketplaces have enabled artisans and tourism businesses to reach customers beyond their local markets. However, the growing availability of products online has also intensified competition, making data-driven business decisions more important than ever.

Many MSMEs still rely on traditional methods for pricing, product selection, and customer engagement. Limited access to market analytics often results in inefficient pricing strategies, inadequate understanding of customer preferences, and missed opportunities for business growth. This creates a strong need for affordable Business Intelligence solutions that help small businesses analyse market trends and make informed strategic decisions.

3.2 Market Trends

Several trends indicate that Business Intelligence is becoming increasingly valuable for MSMEs:

- Rapid growth of e-commerce platforms has increased competition among sellers.
- Customers increasingly compare prices, ratings, and reviews before purchasing products.
- AI-assisted analytics tools have become more accessible and affordable for small businesses.
- Government initiatives promoting digital adoption encourage MSMEs to modernize their operations.
- Businesses are shifting from intuition-based decisions to evidence-based strategies supported by data.

These trends create favourable conditions for adopting Business Intelligence solutions such as **HimalayaInsight**, which helps businesses interpret market information and improve decision-making.

3.3 Competitive Landscape

Although several online platforms support handicraft and tourism businesses, their primary objective is facilitating transactions rather than providing analytical decision support.

Platform	Primary Purpose	Limitation	HimalayaInsight Advantage
Amazon Karigar	Online marketplace for artisans	Focuses on product sales rather than business analytics	Provides pricing analysis, KPI tracking, and market insights
Etsy	Global handmade products marketplace	Limited decision-support capabilities	Offers data-driven business intelligence and strategic recommendations
IndiaMART	B2B marketplace	Primarily supplier discovery	Supports analytical decision-making through dashboards and AI-assisted insights

Unlike these platforms, **HimalayaInsight is not designed as an e-commerce marketplace**. Instead, it functions as a **Business Intelligence and Decision Support System**, enabling MSMEs to analyse pricing patterns, customer ratings, product categories, and market trends through interactive dashboards and AI-assisted interpretation.

The project addresses an important gap by empowering businesses to make informed decisions based on data rather than assumptions, ultimately improving competitiveness and supporting sustainable business growth.

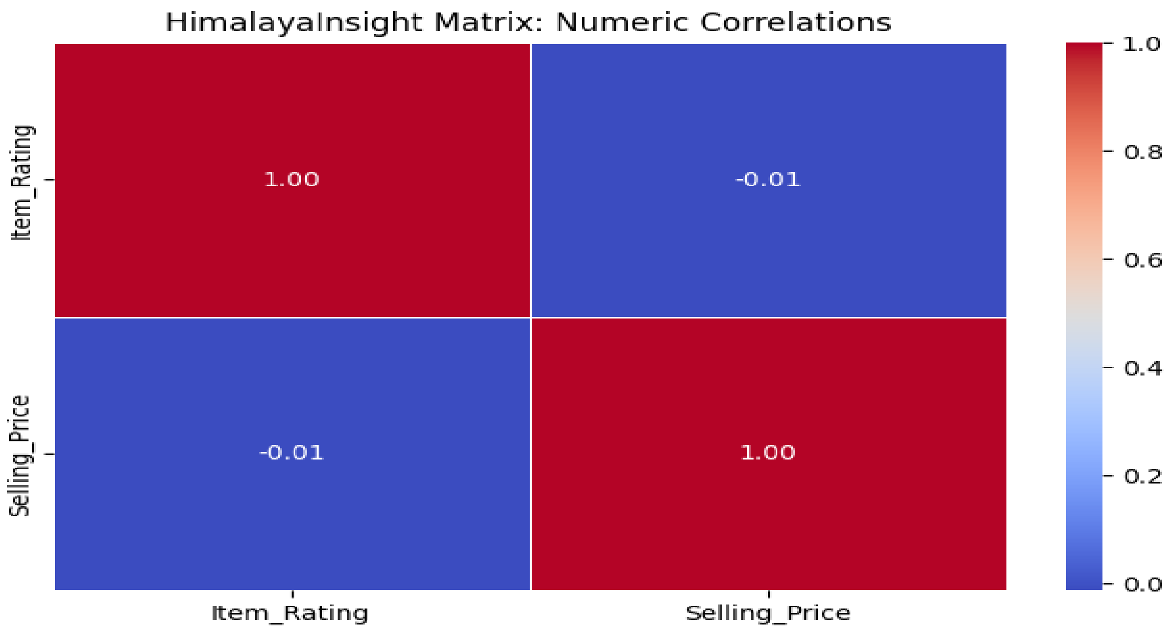

Figure 3. Competitive comparison of existing market platforms and HimalayaInsight's Business Intelligence capabilities.

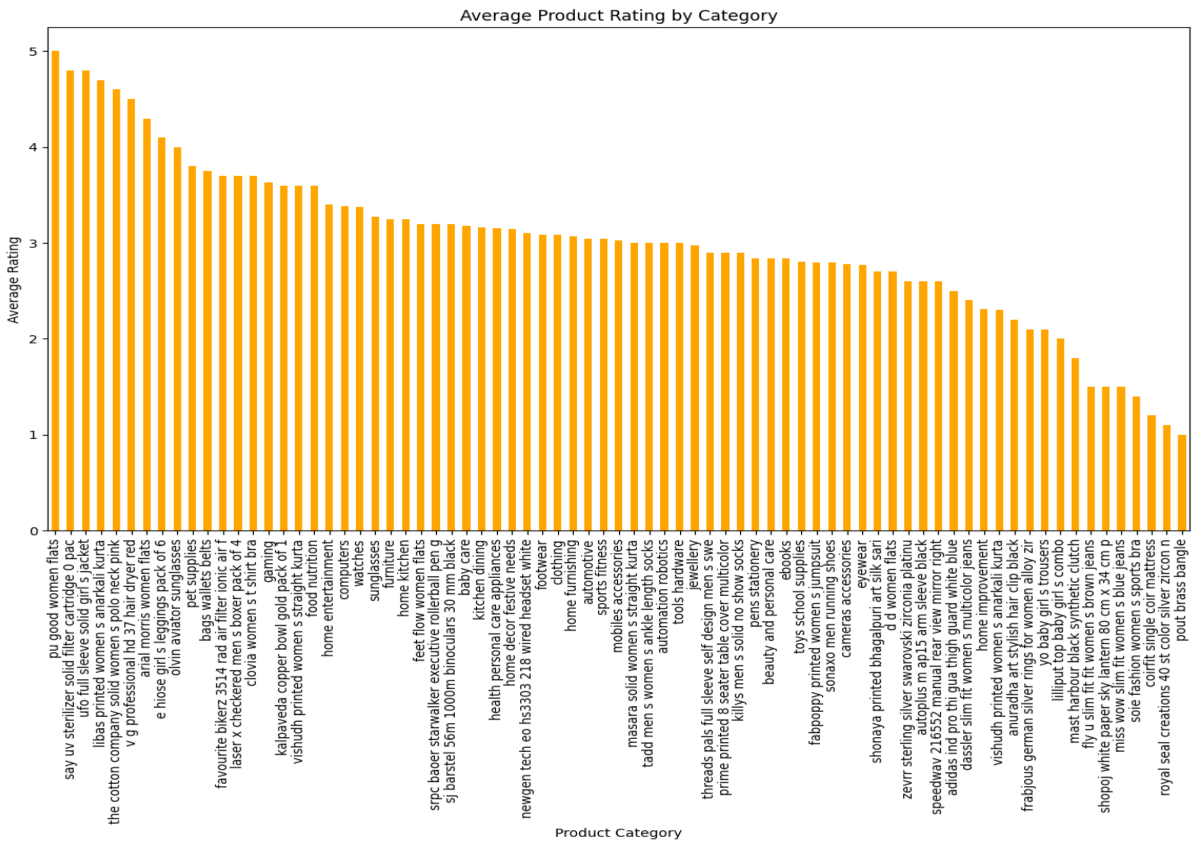

Figure 4. Distribution of products across major categories in the analysed dataset.

Strategy

4.1 Value Proposition

HimalayaInsight is designed to bridge the gap between raw business data and strategic decision-making for Micro, Small, and Medium Enterprises (MSMEs) operating in the handicraft, handloom, and eco-tourism sectors. Many small businesses collect valuable operational and sales data but often lack the analytical capabilities to convert this information into meaningful insights.

The platform simplifies this process by integrating data cleaning, SQL-based analysis, interactive dashboards, and AI-assisted insight generation into a single Business Intelligence workflow. Rather than relying on intuition or fragmented information, business owners can use HimalayaInsight to evaluate pricing trends, monitor customer ratings, compare product categories, and identify emerging opportunities within the market.

The core value proposition of HimalayaInsight lies in enabling affordable, accessible, and data-driven decision-making. By presenting complex analytical findings through intuitive dashboards and actionable recommendations, the platform empowers MSMEs to improve operational efficiency, strengthen competitive positioning, and support long-term business growth.

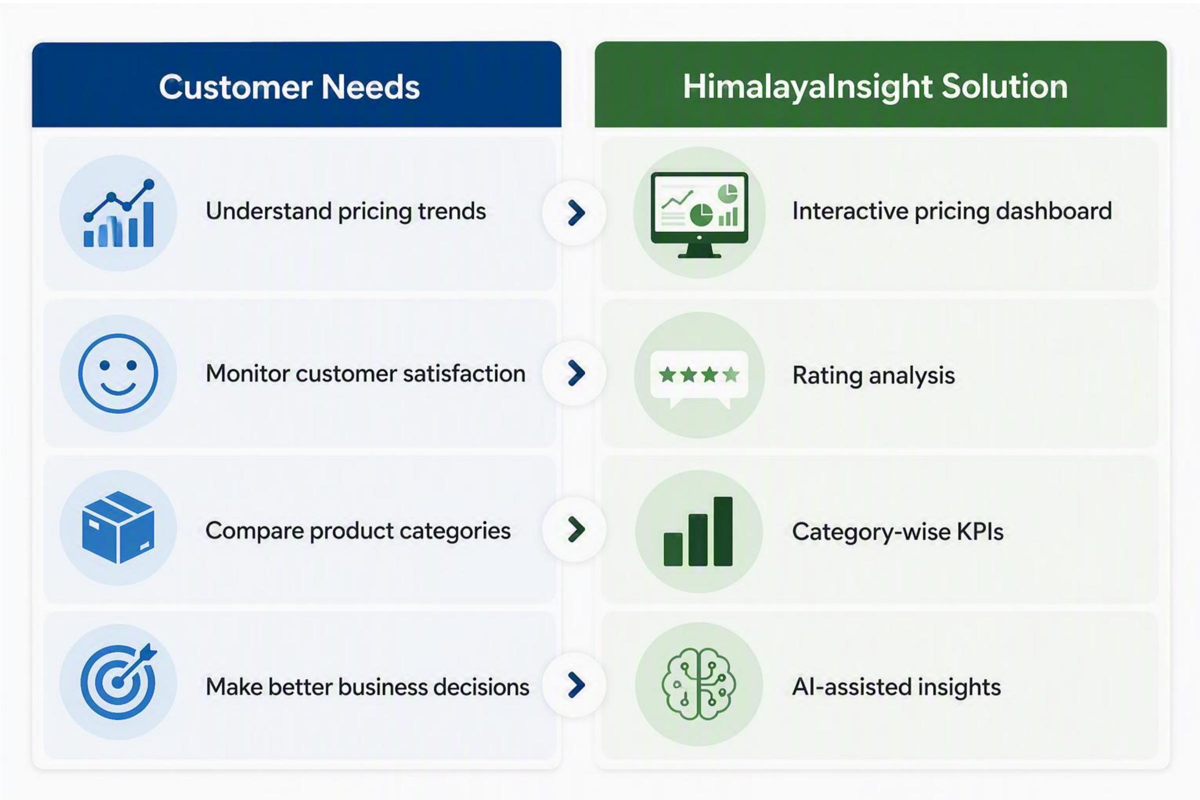

Figure 5. Value proposition of HimalayaInsight for MSMEs.

4.2 Target Customer

The primary users of HimalayaInsight are MSMEs involved in handicrafts, handloom products, and eco-tourism services. These businesses typically have limited access to dedicated analytics teams and require simple yet effective tools for monitoring business performance.

Potential users include:

- Local handicraft manufacturers and artisan groups.
- Eco-tourism operators and homestay businesses.
- Small retailers selling products through e-commerce platforms.
- Startup founders developing rural business initiatives.
- Business consultants supporting MSME growth.
- Government agencies and incubation centres working with local enterprises.

The solution is designed for users who require actionable insights without investing in expensive enterprise Business Intelligence platforms.

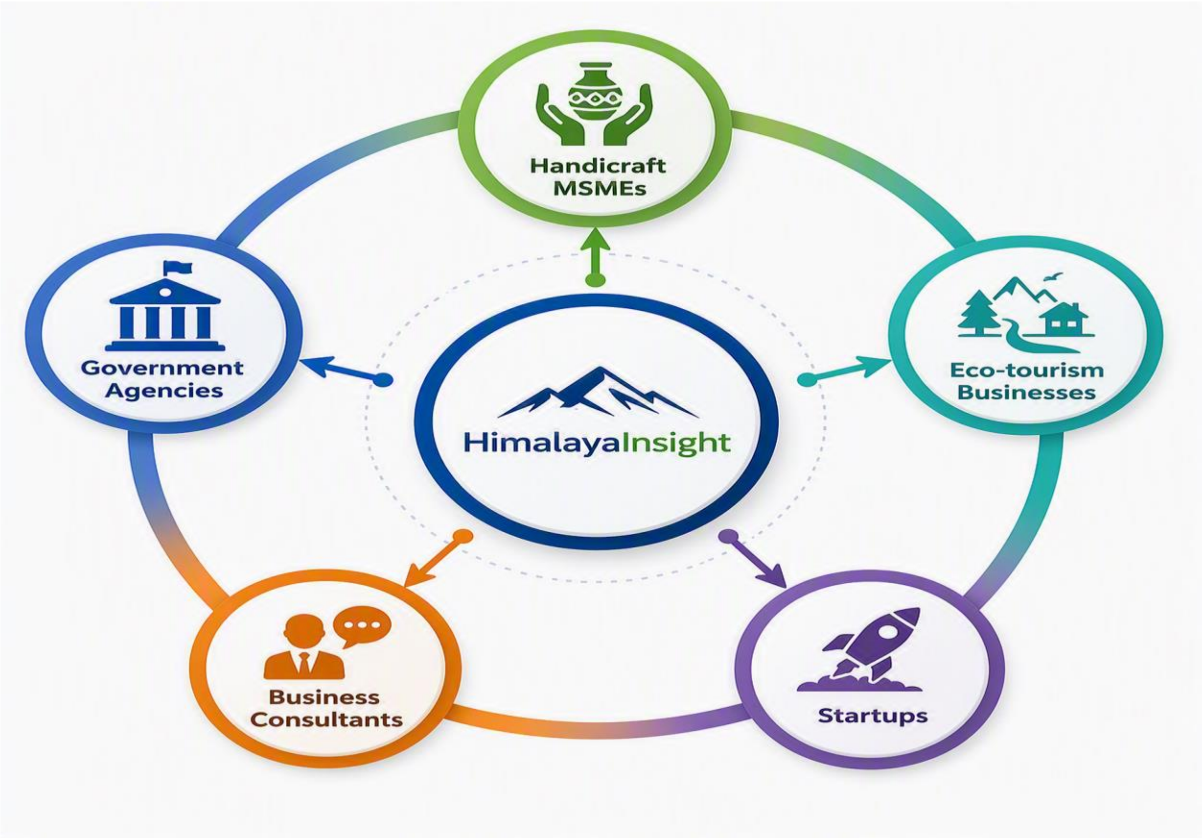

Figure 6. Primary target users of the HimalayaInsight platform.

4.3 Business Development Opportunities

HimalayaInsight has the potential to support multiple business development opportunities beyond its current prototype.

Possible opportunities include:

- Subscription-based access to customised Business Intelligence dashboards.
- Data analytics consulting services for MSMEs.
- AI-powered market research and pricing analysis.
- Dashboard development services for small businesses.
- Training programmes on Business Intelligence and data-driven decision-making.
- Partnerships with incubation centres, entrepreneurship cells, and government-supported MSME initiatives.

These opportunities create multiple revenue streams while increasing the adoption of Business Intelligence practices among small and medium enterprises.

4.4 Outreach Strategy

To encourage adoption, HimalayaInsight should follow a phased outreach strategy.

Phase 1 – Awareness

- Present the solution through university innovation events, startup showcases, and entrepreneurship workshops.
- Publish project demonstrations on LinkedIn and professional networking platforms.
- Conduct introductory webinars highlighting the benefits of Business Intelligence for MSMEs.

Phase 2 – Pilot Implementation

- Collaborate with selected handicraft businesses and eco-tourism enterprises to test the platform.
- Collect user feedback and refine dashboard functionality based on practical business requirements.

Phase 3 – Expansion

- Partner with MSME support organisations, incubation centres, and government agencies.

- Introduce subscription plans and consulting services for businesses seeking advanced analytical support.
- Continuously enhance the platform by integrating additional datasets and AI-powered predictive capabilities.

This phased strategy enables gradual adoption while ensuring that future development remains aligned with the practical needs of the target users.

Figure 7. Proposed business development and outreach roadmap for HimalayaInsight.

Operations

5.1 Operational Workflow

The operational workflow of HimalayaInsight was designed to transform raw business data into meaningful insights that support strategic decision-making. The project follows a structured Business Intelligence process, beginning with data collection and ending with actionable recommendations for MSMEs.

The first stage involved collecting a publicly available e-commerce product dataset from Kaggle. The dataset contained information related to product names, categories, brands, selling prices, market prices, customer ratings, number of ratings, and product descriptions. Before analysis, the dataset was examined for missing values, duplicate records, and inconsistent data types. Data cleaning was performed using Python (Pandas) to ensure accuracy and reliability.

After preprocessing, SQL queries were used to explore the dataset by filtering records, grouping categories, calculating averages, and identifying important business patterns. Exploratory Data Analysis (EDA) was then conducted using Python to generate descriptive statistics and visualisations, allowing a deeper understanding of pricing trends, customer behaviour, and product performance.

The processed data was imported into Tableau Public, where an interactive dashboard was developed. The dashboard included KPI cards, category-wise comparisons, pricing analysis, customer rating summaries, and interactive filters to enable users to explore the data dynamically. Finally, Generative AI tools were used to assist in generating business insights and narratives. Every AI-generated recommendation was verified against the actual analytical results before being included in the project.

This structured workflow ensured that every recommendation presented by HimalayaInsight was supported by reliable data and aligned with Business Intelligence best practices.

Figure 8. Operational workflow followed during the development of HimalayaInsight.

5.2 Standard Operating Procedure (SOP) Summary

The project followed a clear sequence of activities to maintain consistency throughout the analysis:

1. Identify the business problem and define project objectives.
2. Select an appropriate dataset relevant to the business domain.
3. Import the dataset into Python and perform data cleaning.
4. Conduct Exploratory Data Analysis to understand data patterns.
5. Execute SQL queries to answer business-related questions.
6. Create visualisations and dashboards using Tableau Public.
7. Generate AI-assisted business insights and validate them using actual data.
8. Prepare reports, presentations, and documentation for business communication.

Following this standard workflow improved the quality, accuracy, and reproducibility of the analysis.

5.3 Operational Challenges and Improvements

During the project, several operational challenges were encountered. Initially, the dataset required preprocessing because of inconsistencies such as missing values and duplicate records. These issues were addressed through systematic data cleaning and validation using Python.

Another challenge involved selecting appropriate visualisations that clearly communicated business insights. This was resolved by applying suitable chart types and dashboard design principles, ensuring that each visualisation answered a specific business question.

The integration of Generative AI also required careful validation. Some AI-generated observations were broad or lacked sufficient evidence, making it necessary to compare every claim with the actual dataset before including it in reports or presentations. This validation process improved the credibility and reliability of the final business recommendations.

Overall, the project demonstrated that a structured operational workflow, combined with careful data validation and AI-assisted analysis, can significantly enhance the quality of Business Intelligence solutions.

Financial Model

6.1 Revenue Model

HimalayaInsight is designed as a Business Intelligence platform that supports MSMEs by providing data-driven insights through dashboards and AI-assisted analysis. A subscription-based Software-as-a-Service (SaaS) model is proposed as the most suitable approach for future implementation. This model offers affordable access to analytical dashboards while generating recurring revenue.

Potential revenue sources include:

- Monthly or annual dashboard subscriptions
- Custom dashboard development
- Business Intelligence consulting
- AI-assisted market analysis
- Corporate training and workshops

Diversifying these services can help ensure long-term sustainability while addressing the needs of businesses of different sizes.

Figure 9. Proposed revenue model for HimalayaInsight.

6.2 Proposed Pricing Structure

Plan	Target User	Suggested Price
Basic	Individual MSMEs	₹499/month
Professional	Growing Businesses	₹999/month
Enterprise	Organisations & Incubators	Custom Pricing

The above pricing is indicative and included only as a conceptual business model for this capstone project.

Project Plan & Risk

7.1 Project Timeline

The HimalayaInsight project was completed through a structured nine-week internship, with each module building upon the previous one. The project began with identifying a business problem and selecting a suitable dataset. This was followed by data cleaning, SQL analysis, exploratory data analysis, dashboard development, AI-assisted insight generation, and business storytelling. The final weeks focused on preparing the capstone report, validating AI-generated outputs, and presenting strategic recommendations. This phased approach ensured steady progress while allowing continuous refinement of the project deliverables.

7.2 Key Risks and Mitigation Strategies

Risk	**Potential Impact**	**Mitigation Strategy**
Poor data quality	Inaccurate insights and unreliable analysis	Perform data cleaning, validation, and consistency checks before analysis.
Limited adoption by MSMEs	Reduced practical impact of the solution	Design simple dashboards, provide user training, and gather feedback through pilot testing.
Dependence on static datasets	Insights may not reflect current market conditions	Integrate updated datasets or real-time data sources in future versions of the platform.

Identifying these risks early helped maintain the quality of the analysis and ensured that recommendations remained realistic and evidence-based.

Strategic Recommendations

Recommendation 1: Use Data-Driven Product Strategy

Insight: Product demand and customer satisfaction vary significantly across categories based on ratings, reviews, and market performance.

Recommendation: MSMEs should analyze customer preferences and market trends before launching new products. Focus should be given to high-demand categories and products with strong customer acceptance.

Expected Impact: Better product selection, reduced market risk, and improved customer satisfaction.

Recommendation 2: Optimize Pricing Through Market Intelligence

Insight: Competitive pricing combined with strong product value influences customer purchase decisions.

Recommendation: Businesses should regularly monitor competitor pricing, customer response, and product positioning to develop effective pricing strategies. Premium products should emphasize uniqueness, craftsmanship, and authenticity.

Expected Impact: Improved profit margins, stronger brand value, and sustainable revenue growth.

Recommendation 3: Adopt AI-Powered Business Intelligence

Insight: Data visualization and analytics can convert raw market data into actionable business insights.

Recommendation: MSMEs should adopt digital intelligence tools for tracking trends, analyzing customer feedback, monitoring competitors, and identifying growth opportunities.

Expected Impact: Faster decision-making, increased digital adoption, and scalable business growth.

Conclusion

By adopting data-driven strategies, smart pricing approaches, and AI-enabled analytics, Uttarakhand MSMEs can improve competitiveness, expand market reach, and build sustainable growth models.

Limitations & Next Steps

9.1 Limitations

Although the analysis provides valuable insights into product trends, pricing patterns, and customer preferences, certain limitations exist:

- **Dataset Limitation:** The analysis is based on available e-commerce product data and may not represent the complete market scenario of all Uttarakhand MSMEs.
- **Limited Business Variables:** Factors such as production cost, supply chain efficiency, seasonal demand, and offline customer behavior were not included.
- **Market Validation:** Recommendations are based on analytical findings and require real-world validation through MSME feedback and market testing.
- **Data Availability:** Continuous access to updated market and customer data is required for maintaining accurate insights.
- **External Factors:** Economic changes, competitor strategies, and policy changes may influence business outcomes.

9.2 Next Steps

To enhance the capabilities and practical application of the project, the following improvements can be considered:

- **Real-Time Data Integration:** Connect live marketplace, tourism, and customer review data sources for continuous analysis.
- **AI-Based Prediction Models:** Implement machine learning models for demand forecasting and future trend prediction.
- **MSME-Specific Dashboard:** Develop customized dashboards for different business sectors such as handloom, handicrafts, and eco-tourism.
- **Sentiment Analysis Enhancement:** Use Natural Language Processing (NLP) to analyze customer reviews and identify improvement areas.
- **Field Validation:** Collaborate with local MSMEs to test recommendations and measure business impact.

Appendices

Appendix A: Competitive Matrix (Week 2)

The competitive analysis was conducted to understand existing market intelligence solutions and identify opportunities for HimalayaInsight.

Platform	Key Features	Strengths	Limitations
Google Trends	Keyword and search trend analysis	Free and easy market trend discovery	Limited business-specific insights
Tableau Public	Data visualization dashboards	Interactive analytics and reporting	Requires prepared datasets
Power BI	Business intelligence and reporting	Advanced visualization capabilities	Requires technical knowledge
Marketplace Analytics Tools	Product and competitor tracking	Useful for e-commerce insights	Often expensive for small businesses
HimalayaInsight	AI-powered rural business intelligence	Localized insights for Uttarakhand MSMEs	Requires further real-world validation

Appendix B: Revenue Model (Week 6)

Revenue Model Link:
https://public.tableau.com/views/HimalayaInsightDashboard/Dashboard1?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Proposed Revenue Streams:

1. **Subscription-Based Model**
   a. Monthly/annual plans for MSMEs to access analytics dashboards and market insights.
2. **Premium Consulting Services**
   a. Customized business recommendations, market analysis, and growth strategies.
3. **Data Intelligence Reports**
   a. Industry-specific reports on trends, pricing, and customer preferences.
4. **Partnership Model**
   a. Collaboration with government organizations, tourism boards, and MSME development programs.

Appendix C: Standard Operating Procedure (SOP) (Week 4)

SOP Link:

https://colab.research.google.com/drive/1EOpE39llgx26TK4h260CUL2Ae9XTVAdk?usp=sharing

Process Workflow:

1. **Data Collection**
    a. Gather product, pricing, customer review, and market trend data.
2. **Data Cleaning**
    a. Remove inconsistencies, missing values, and duplicate records.
3. **Data Analysis**
    a. Perform exploratory data analysis to identify trends and patterns.
4. **Dashboard Development**
    a. Create visual reports using BI tools such as Tableau/Power BI.
5. **Insight Generation**
    a. Convert analytical findings into actionable business recommendations.
6. **Decision Support**
    a. Provide strategic recommendations for MSME growth.

Appendix D: Pitch Deck (Week 8)

Pitch Deck Link:
https://youtu.be/Dmm9nobTHHo

The pitch deck presents:

- Problem Statement
- Market Opportunity
- Proposed Solution: HimalayaInsight
- Data Analysis Approach
- Key Insights
- Business Model

- Strategic Recommendations
- Future Scope

Appendix E: References

1. Kaggle – E-commerce Product Dataset
   URL: https://www.kaggle.com/
2. Tableau Public – Data Visualization Platform
   URL: https://public.tableau.com/
3. Google Trends – Market Trend Analysis Tool
   URL: https://trends.google.com/
4. Ministry of Micro, Small & Medium Enterprises (MSME), Government of India
   URL: https://msme.gov.in/
5. Uttarakhand Tourism Development Board
   URL: https://uttarakhandtourism.gov.in/